Access to healthcare, fertility, and reproductive health

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We examine the impact of excluding unauthorized immigrants from public healthcare on fertility behaviors and birth-related outcomes by exploiting a reform in Spain in 2012. Using comprehensive administrative data on the universe of births and abortions, we implement a synthetic difference-in-differences design to compare outcomes across nationalities differentially affected by the 2012 reform. We find a fall in abortion rates and an increase in birth rates among women from countries with higher shares of unauthorized immigrants following the policy change, suggesting restricted access to reproductive healthcare. We also show that births from immigrant women affected by the reform were more likely to involve complications, worsening maternal health, in part due to changes in the composition of mothers. Our findings underscore the unintended consequences of policies that restrict access to healthcare and their effects on prenatal care usage, contraceptive access, and long-term maternal and child health.

Keywords: Fertility, abortion, neonatal health, healthcare

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1 Introduction

Fertility control plays a crucial role in shaping women's lives, influencing labor force participation (Goldin and Katz, 2002; Bailey, 2006), earnings and educational attainment (Bailey, 2006; Ananat and Hungerman, 2012), and even exposure to domestic violence (Deschamps, 2024). Understanding the determinants of fertility decisions is therefore central for both economic and public health research. In this paper, we study how restricting access to public healthcare affects reproductive outcomes for immigrant women. We exploit a 2012 reform in Spain that excluded unauthorized immigrants from the public healthcare system and examine its effects on abortions, births, and birth-related health outcomes.

Using administrative data covering the universe of births and voluntary pregnancy terminations in Spain, we implement a synthetic difference-in-differences design that compares women from nationalities with high and low shares of unauthorized immigrants. Our results show that the reform had substantial effects on fertility behavior. Women from nationalities in the top quartile of the unauthorized share were approximately 23% less likely to undergo an abortion and about 10% more likely to give birth relative to women from nationalities in the bottom quartile. We also find evidence of deteriorating delivery conditions: women more exposed to the reform experienced higher rates of birth complications and caesarean deliveries.

The relationship between access to healthcare and fertility decisions has been studied in several contexts. Expanding access to long-acting reversible contraceptives has been shown to reduce unintended pregnancies and lead to declines in both births and abortions among low-income women (Kelly et al., 2020). Conversely, restricting access to abortion tends to increase births (Venator and Fletcher, 2021) and may worsen infant health outcomes, including higher infant mortality rates (Bell et al., 2025; Gemmill et al., 2024). These findings highlight the importance of reproductive healthcare access for fertility control.

Access to healthcare during pregnancy is also closely linked to infant health through the Fetal Origins Hypothesis (Barker, 1998), which emphasizes that in-utero conditions can have

long-lasting consequences for health and development. A large body of work shows that prenatal care improves birth outcomes such as birth weight and neonatal survival. Evidence from Thailand and Ghana indicates that expanding healthcare access and reducing financial barriers increases prenatal care utilization and lowers infant mortality (Gruber et al., 2014; Osei et al., 2025), with similar associations documented in sub-Saharan Africa (Engdaw et al., 2023) and China (Wenling et al., 2024). However, findings are not uniform: Dave et al. (2019) report that expanding healthcare access for low-income families in the United States did not improve—and in some cases worsened—birth outcomes.

A growing literature examines how healthcare policies affect immigrant populations in particular. Extending healthcare coverage to immigrants has generally been shown to improve healthcare utilization and neonatal outcomes (Miller et al., 2022; Kose et al., 2024), though evidence from Canada suggests more limited effects (Wilson-Mitchell and Rummens, 2013). Other studies highlight the importance of the broader policy environment: immigrant-friendly policies improve birth outcomes, while immigration enforcement and policy uncertainty can worsen them through stress and reduced healthcare utilization (Timilsina, 2023; Vu, 2024; Torres et al., 2022). Despite the importance of prenatal care, immigrants often underutilize these services (Boerleider et al., 2013), and the "healthy immigrant paradox" suggests that immigrant mothers initially experience better birth outcomes than native-born women despite socioeconomic disadvantage (Markides and Coreil, 1986) -an advantage that tends to diminish with longer residency (Giuntella, 2016). Together, these findings suggest that the relationship between healthcare access and maternal health among immigrant women is complex.

Most existing evidence comes either from developing countries or from the fragmented healthcare system of the United States. There is relatively little causal evidence from high-income European countries with universal healthcare systems, where both the institutional context and the consequences of exclusion may differ substantially. Spain provides a particularly informative case. In 2012, the Spanish government implemented a reform that removed healthcare access for unauthorized immigrants, shifting the system from universal coverage to a model based on social security contributions. Although pregnancy-related care

was formally exempted, unauthorized immigrants may still have avoided healthcare services due to fear of deportation, lack of information about their rights, or administrative barriers. This reform provides a natural experiment to study the consequences of removing healthcare access in a universal healthcare system. We examine its effects using administrative data covering all births and abortions in Spain between 2010 and 2015. To proxy unauthorized status, we follow Mestres et al. (2021) and use the share of unauthorized immigrants by nationality. Using a synthetic difference-in-differences framework, we compare outcomes across women from nationalities with different exposure to the reform. Previous work found that the reform reduced planned care (Jiménez-Rubio and Vall Castello, 2020), increased mortality rates (Mestres et al., 2021), and increased the probability of experiencing domestic violence (Bellés Obrero et al., 2023).

Our results indicate that restricting healthcare access significantly altered fertility behavior and delivery outcomes. In addition to the large decline in abortions and increase in births, we find evidence that women more exposed to the reform experienced higher rates of delivery complications and caesarean sections. These patterns may reflect reduced access to prenatal care, but also changes in the composition of mothers: after the reform, new mothers from the most affected nationalities were more likely to be over 35 and less likely to have a partner, characteristics associated with higher delivery risk. We do not find heterogeneous effects by regional implementation intensity, but we provide suggestive evidence that foreign women residing in regions where the reform was implemented more strictly were less likely to report using the contraceptive pill after the reform.

Overall, our findings contribute to the literature on healthcare access, fertility control, and immigrant health by providing causal evidence from a high-income country with a universal healthcare system. They highlight how policies restricting healthcare access—even when exemptions formally exist—can have substantial consequences for reproductive outcomes among vulnerable populations.

The paper proceeds as follows. Section 2 presents the context and the 2012 healthcare

reform, Section 3 explains our identification strategy, Section 4 describes the data, and Section 5 reports our results, including a discussion of potential mechanisms, robustness checks, and additional results. Section 6 concludes.

2 Context

2.1 The 2012 healthcare reform

The Spanish healthcare system was characterized by universal access, including immigrants regardless of their legal status, since the 1908s. Funded primarily through taxes, the system ensured that residents had access to health services without direct contribution requirements. In 2000, it was ranked as the 7th most efficient healthcare system in the world by the World Health Organization (2000). To access health services, immigrants, including those without formal residency status, needed to register with local authorities in their place of residence. By presenting a valid ID and proof of residence, they could obtain a local registration document that granted access to public healthcare and education (García-Armesto et al., 2010). This inclusive approach was designed to promote public health and safety by reducing untreated illnesses that could potentially affect the broader community. However, the system faced financial sustainability challenges, exacerbated by the 2008 economic crisis, leading to rising concerns over costs.

In 2012, Spain, as part of a set of "austerity" measures, implemented significant budget cuts in healthcare (Legido-Quigley et al., 2013). The cuts were accompanied by Royal Decree-Law 16/2012, approved on April 20, 2012, which significantly restructured access to healthcare. A second decree followed this reform in August 2012, the Royal Decree-Law 1192/2012, which further clarified and expanded on the provisions set out in the initial reform, focusing on specific regulatory adjustments and further defining the scope of beneficiaries and their eligibility criteria under the revised system. The law came into effect on the 1st of September. Healthcare access was restricted to individuals registered in the Spanish social security system through employment or long-term residency contributions. Exceptions were

made for certain vulnerable groups: pregnant women, children, and emergency care in case of accidents (Lopez-Valcarcel and Barber, 2017).

After the reform was implemented, the healthcare cards of individuals who did not qualify for health, unauthorized immigrants in particular, were electronically canceled. Government reports show that 873,000 healthcare cards had been canceled by September 2012 (about half a million from people no longer in the country, 160,000 from unauthorized migrants, and 200,000 from EU citizens not entitled to healthcare under the new rules) (Durana and Moreno-Fuentes, 2016).

Despite the national mandate, the implementation of the law varied across different regions of Spain (Cimas et al., 2016). Some regions developed their own programs to provide healthcare for unauthorized immigrants. Notably, four regions (Asturias, Basque Country, Galicia, and Catalonia) had already put in place alternative measures before the law's enforcement. These measures varied in strictness, with Asturias establishing the least stringent requirements for access, while the Basque Country had the highest. On the other side, six regions (Madrid, Murcia, Balearic Islands, Castile-Leon, La Rioja, and Castile-La Mancha) did not approve almost any other accompanying measure, so that the reform was applied in full effect. All the other regions (autonomous communities) implemented alternative policies by the end of 2013, with different degrees of requirements and coverage.

In July 2018, Spain reinstated universal healthcare access with the approval of Royal Decree-Law 7/2018, which recognized access to the National Health System as a fundamental right of every person in Spain.

2.2 Fertility and immigration in Spain

In 2024, Spain had 48,592,909 registered residents, of which 6,491,502 (about 13%) with a foreign nationality. According to the Spanish National Institute of Statistics (INE), migrants play a crucial role in the country's demography. For instance, most of the population growth recorded in the last quarter of 2023 was due to the rise in people born abroad who acquired citizenship (Instituto Nacional de Estadística (INE), 2024). A sizeable share of

immigrants reside in the country illegally: between a fifth and a fourth of the total does not have a legal permit to reside in Spain. The number of individuals registered with the municipalities declined starting in 2011 and sharply dropped since 2013, finally stabilizing in 2017 and increasing again in 2018. Figure A2 in the Appendix, reports the total number of municipal records for individuals in our population of interest (whose countries of nationality have not ratified the Schengen agreements) in our period of interest (2010-2015).

According to a recent working paper (Gálvez-Iniesta, 2020), four out of five unauthorized immigrants are under the age of 40, and when focusing on men, over 70% are younger than 30 years old. unauthorized immigrants are, on average, younger than other immigrants. Women represent 55% of the irregular immigrant population in Spain. In line with our own estimates, over three-fourths of the unauthorized residents come from Latin America.

Immigrants contribute significantly to the Spanish birth rate, with immigrants from the Maghreb from the first generation displaying a fertility rate almost double the Spanish one. There is, however, quite some heterogeneity, with first-generation individuals from Latin America having fewer children than locals (García-Gómez et al., 2023).

Spain has undergone significant demographic and legal changes in reproductive health over the past decades, with abortion laws evolving alongside a shifting birth rate driven by immigration. The most crucial reform occurred in 2010. The 2010 reform established abortion as a right, allowing it on demand up to 14 weeks of pregnancy and up to 22 weeks in cases of severe fetal abnormalities or health risks. In recent years, additional changes have strengthened access, such as the 2022 reform that removed parental consent requirements for 16- and 17-year-old women. Despite this, disparities remain between native and immigrant women, with higher abortion rates among immigrants, likely due to economic insecurity, lower contraceptive access, and/or legal or linguistic barriers (Mandroiu et al., 2024). Figure A3 in the Appendix reports the number of births and abortions over time for women with non-Schengen nationality.

3 Identification Strategy

Our identification strategy is predicated on the assumption that unauthorized immigrants were more impacted by the policy than legal residents. As the available data sources do not reveal immigration status, we approximate legal status using nationality. Following the approach of Jiménez-Rubio and Vall Castello (2020), we posit that women from nationalities with a higher proportion of unauthorized migrants in 2011 are more likely to be unauthorized and thus more affected by the policy compared to those from nationalities with lower proportions of unauthorized migrants in Spain. We use the 2011 figures for the share of unauthorized immigrants to mitigate any potential effects of the policy change, thus adhering to an intention-to-treat (ITT) approach.

To identify the share of unauthorized immigrants by nationality, we use public data about the number of residence permits and the number of people registered with the municipalities in Spain by nationality. The share of unauthorized immigrants is computed by dividing the estimated number of residents with no permits (which we proxy with the total number of individuals registered with the municipalities minus the total number of visa permits), by the total number of residents in a given year. We consider 2011 as the year of reference. Our estimated fraction of unauthorized immigrants varies substantially across nationalities, ranging from Paraguay and Honduras, recording over 50%, to Morocco and China, whose citizens almost always have a residence permit. The full list of nationalities and shares can be found in Table A1, in the Appendix[1]. We restrict our analyses to women whose nationality is not covered by the Schengen agreements, as these agreements extend healthcare coverage to all member states.

Figure 1 reports the distribution of the share of unauthorized immigrants in 2011 at the country level (left) and the individual level (right) for all the immigrant women who gave birth in Spain. These two plots show the likelihood that a given observation (a country, in

[1]We repeated the same exercise at the Regional level (see Figure A4 in the Appendix), but some nationalities display a negative share of unauthorized immigrants (i.e., the number of individuals with a visa is larger than those registered in the Padron), which flags that probably individuals moved across provinces without updating their residence. Therefore, we keep the analyses at the country level to reduce the noise.

the first case, and an individual, in the second) is "treated". At the individual level, this represents the estimated probability of being an unauthorized resident.

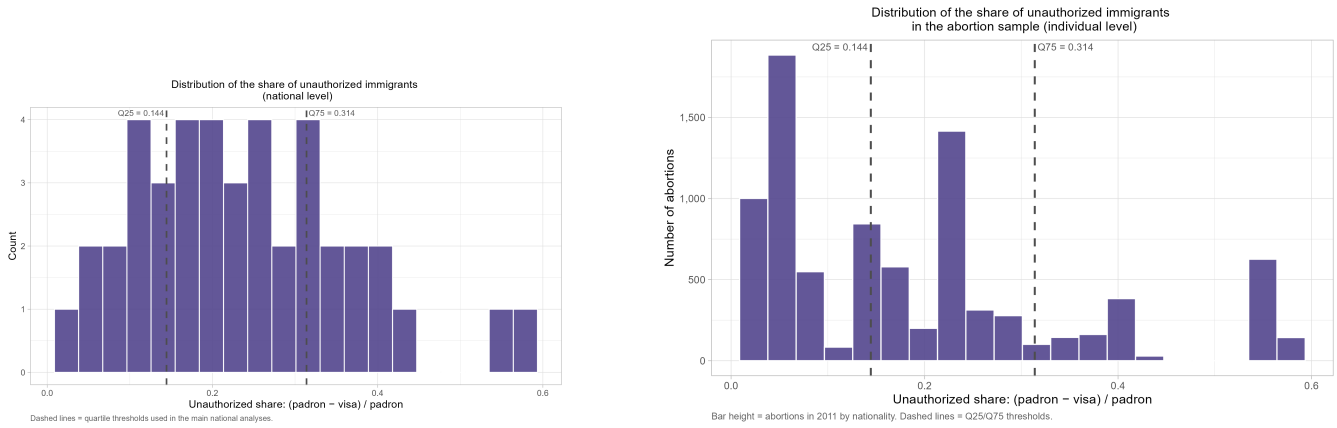


Figure 1: Histograms reporting the fraction of unauthorized immigrants from the women's country of origin. In the left panel, the figure reports the percentage of countries with the share of unauthorized immigrants in 2011 in each bin. In the left panel, the units are women having an abortion in 2011.

As shown in the plots, a substantial number of registered immigrants (over 25%) come from countries (mainly Morocco and China) with almost no unauthorized immigrants (the first quartile is equal to 0.02), while about a quarter of the residents have a nationality with over 25% of immigrants with no residence permit (the third quartile is 0.23). We use as a continuous "treatment" the share of unauthorized immigrants for a given nationality, as an approximation of the probability that a woman of the same nationality resides in Spain illegally.

Our baseline model is a two-way fixed-effects model with panel data, in which the units are the countries of nationality and time is expressed in quarters. We consider two different "first" periods for the treatment according to the outcome evaluated. When looking at abortions and birth-related events, we assume that the effects of the policy are quite immediate, so the first treated period is the quarter right after the policy adoption (the last quarter of 2012). When instead we evaluate the number of births and the mother's characteristics, we believe the effects of the reform to be detectable only at least 6 months after the reform implementation, given that pregnancies last 9 months and abortions are extremely rare after 14 weeks. Hence, in this case, we consider the second quarter of 2013 as the first treated

period[2]. We have two sets of outcomes: some at the country of origin level, and some at the individual level. Our main outcomes of interest are birth and abortion rates by nationality.

$$Outcome_{ct} = \alpha + \sum_{t=2010_1}^{2015_4} \beta_t \; quarter_t \times Share.unauthorized.2011_c + \sum_{t=2010_1}^{2015_4} \gamma_t \; quarter_t + \sum_{c=1}^{42} Country_c + \epsilon_{ct}$$

where the set of β_t represents the coefficients of interest, and $quarter_t$ and $Country_c$ represent the time and observation fixed effects.

Our second set of outcomes (at the individual level) includes birth-related outcomes and women's characteristics. The specification at the individual level is similar, but we add Province $\times$ Country of origins fixed effects to control for spatial patterns of immigration (and hospital quality). We weigh observations by the number of registered residents in 2011 from each nationality to give a larger weight to countries with a larger population. We always cluster standard errors at the fixed effects level: in the first case, the level of clustering is the country of origin, while in the second case, it is the country of origin $\times$ Region.

Given that the pre-trend assumption does not hold clearly for one of our main outcomes (the number of births), we choose to implement a synthetic difference-in-differences (Arkhangelsky et al., 2021). This methodology extends standard DiD by combining unit and time weights. On the unit side, it constructs a synthetic control to better match the treated units' pre-treatment trajectory. On the time side, it assigns weights across pre-treatment periods according to their predictive power for post-treatment outcomes in the control group. These time weights are then applied to both treated and control units, so that comparisons

[2]In this case, the first treated period would be after seven months, instead of six. Hence, women in the third trimester who would have wanted to have an abortion in September but could not, would still be in the control group. However, over 75% of abortions occur within the first two months of pregnancy (see Figure A5 in the Appendix), hence women in this group would be relatively rare. Furthermore, if anything, having a (small) share of treated women within the control group would bias our estimates downward.

emphasize periods with genuine forecasting content while down-weighting noisy or uninformative segments of the pre-treatment history. To avoid overfitting, the weights are estimated in a penalized fashion (ridge regularization), which shrinks them toward uniformity and prevents extreme solutions. To compute standard errors, we employ the clustered jackknife resampling technique[3].

The main identifying assumption of the synthetic difference-in-differences is that, once unit and time weights are applied, the weighted average outcome trajectory of the controls provides a valid counterfactual for the treated group in the absence of treatment. This implies that weighted parallel trends hold: after reweighting, the pre-treatment evolution of outcomes in the treated group can be replicated by the control group. A second assumption is that there is no anticipation: treated units should not be affected by the treatment before it occurs. This latter one is supported by the fact that the policy was unexpected and was applied within a few months of its approval. Thirdly, we assume that the Stable Unit Treatment Value Assumption (SUTVA) holds, as it is reasonable to exclude the possibility that this policy had no effect on regularized immigrants. Finally, we should be mindful of compositional changes in the population. Mestres et al. (2021) reports that there are some small compositional changes in the population of interest, with more educated immigrants more likely to leave the country. We find some evidence that regions where the law was not strictly enforced saw a mild increase in immigration from the populations most affected by the reform, whereas those in high-implementation regions did not (see Fig. A6 in the Appendix).

As our treatment variable is continuous, we dichotomize it to implement the synthetic difference-in-differences design. In the baseline specification, we flag whether a mother's nationality falls into the top (the share of unauthorized immigrants from her nationality exceeds 31.3%,) or bottom (the share of unauthorized immigrants from her nationality is

[3]This method systematically omits one cluster (here country) at a time from the dataset, re-estimates the model on the remaining $n-1$ observations, and evaluates the statistic of interest for each subsample. This process yields a set of n pseudo-values, whose variability is then used to estimate the standard error. Formally, the jackknife standard error is calculated as the square root of the scaled variance of these pseudo-values, offering a robust, non-parametric measure of uncertainty that does not rely on strong distributional assumptions.

below 14.7%,) quartile with regard to the proportion of unauthorized residents in Spain in 2011, with other cases marked as missing.[4]. Nationalities in the fourth quartile, which have a larger share of unauthorized immigrants, are considered as "treated". This choice maximizes treatment contrast and ensures that the control group is composed of units that are essentially untreated, making the identifying assumptions more credible. As a robustness check, we also consider an alternative splitting of the sample above and below the median. This approach allows us to retain more observations and increase precision, though it may attenuate estimates since the control group includes partially exposed units.

4 Data

We use data for all quarters between 2010 and 2015, from several sources. For abortions, we use the microdata from the administrative registers of the Spanish Ministry of Health. This register contains information about all abortions performed in Spain each year, along with some details about the women receiving the abortions, such as their age, marital status, nationality, and province of residence. Our data includes a total of 177,980 abortions from women from 42 countries[5] (not covered by the Schengen agreements) of origin during 6 years (2010-2015). Table A2 in the Appendix reports the demographics of this population.

For birth rates, we use microdata from birth certificates collected by the Instituto Nacional de Estadística (INE), which covers all births registered in Spain each year. They also contain some demographic information, such as the mother's nationality, age, and marital status, along with data on the delivery, including whether the birth was premature, the baby's weight, and whether the delivery was performed via c-section. The province of birth is also available. We consider a total of 372,123 births of women from 42 countries of origin during 6 years (2010-2015). On average, mothers in the sample are 29 years old. About 4% of the

[4]We keep the same thresholds also for the individual-level analyses, even if the quartiles are different, in order to keep the same countries as "treated" or "control".

[5]The list of countries can be found in Table A1, in the Appendix

sample is composed of teenagers, while 15% of the women are older than 35. Table A3 in the Appendix reports the demographics of the population of interest.

We collapse the data at the country of origin $\times$ quarter level, and obtain two balanced panel samples of 1008 observations each.

We first evaluate the effects of the policy on fertility, namely on births and abortions. We calculate abortion and birth rates by adding up the number of abortions (or births) from women of a given nationality in a given quarter, and dividing this figure by the number of registered co-nationals (regardless of legal status). We multiply by 1000 for ease of interpretation. Table A1 in the Appendix, reports the number of women giving birth and interrupting a pregnancy in 2011 by nationality. A potential problem with our measures of birth and abortion rates is that they are computed using the number of registered immigrants in the denominator, as a measure of the total number of immigrants from a given nationality. However, one of the main reasons for registering with the municipality was access to healthcare, which was ceased after the reform. Thus, the figures coming from the local registry may underestimate the real number of residents without a permit, thus inflating our outcomes.

We then focus on birth-related outcomes: pre-term delivery, underweight newborns ($< 2,500g$), complications during delivery, and cesarean sections. We create binary indicators for each of these outcomes in our birth registry data.

To investigate potential mechanisms, we evaluate the characteristics of new mothers, which may change as a result of changes in pregnancy or abortion rates: we consider women's age, their marital status (we flag women who are not married, nor in a civil union), and whether information about the father is recorded, using this as an indicator for whether the father is known.

Finally, we consider the 2012 Spanish Health Survey (ENSE) and the 2014 European Health Survey (EESE) to investigate contraceptive usage.

5 Results

5.1 Fertility

We begin by examining the effects of the reform on abortion rates. The direction of the effect is a priori ambiguous: the hostile political climate or fear of inadequate care might have prompted women to terminate pregnancies, either by using the pregnancy-related care exemption included in the reform or by paying out of pocket. At the same time, a strict implementation of the reform may have introduced barriers to accessing abortion services, leading to a decline in terminations. Our results support the latter: following the reform, women from nationalities in the top quartile of the unauthorized immigrant share were about 23% less likely to have an abortion (0.5 fewer abortions per 1,000 inhabitants). The TWFE model delivers comparable results: a 29% decrease using the binary treatment, and approximately 0.6 fewer abortions per 1,000 inhabitants for women from a nationality with a 30% unauthorized immigrant share (the third quartile value). Results are presented in Table 1, and dynamic estimates are shown in Figure 2.

The dynamic ATT and pre-trend estimates, reported in Figure A7 in the Appendix, show pre-trends distributed around zero and small in magnitude, indicating that control units were well matched to treated ones. All post-reform ATT estimates are significantly different from zero. Given the relatively small number of nationality groups (11 treated, 11 controls), standard errors are somewhat imprecise; reassuringly, results are consistent across specifications. We also estimate the model on the full sample, comparing nationalities above and below the median unauthorized share (Figure A8 in the Appendix). Results are consistent with the main specification (ATT = -0.505, s.e.=0.171). Additional robustness checks using a standard DiD on the same binary sample (Figure A9) and a continuous treatment on the full sample (Figure A10) confirm the main finding: the reform led to a significant decline in abortions among affected women.

Table 1: Static Difference-in-Differences with the continuous and the binary treatment variable (estimated share of unauthorized immigrants): Share of abortions and births per thousand people.

VARIABLES	(1) Synth DID $\frac{N_Abortions}{Residents}$	(2) TWFE $\frac{N_Abortions}{Residents}$	(3) TWFE $\frac{N_Abortions}{Residents}$	(4) Synth DID $\frac{N_Births}{Residents}$	(5) TWFE $\frac{N_Births}{Residents}$	(6) TWFE $\frac{N_Births}{Residents}$
ATT	-0.56***			0.46**		
	(0.182)			(0.219)		
Share Un. Imm.>3Q*After		-0.705**			0.619**	
		(0.302)			(0.272)	
Share Un. Imm.*After			-2.050**			2.106***
			(0.942)			(0.612)
After		0.104	0.206		-0.117	-0.266
		(0.164)	(0.219)		(0.349)	(0.312)
Constant		2.338***	2.523***		5.326***	5.011***
		(0.0620)	(0.0613)		(0.227)	(0.192)
Observations	528	528	1,008	528	528	1,008
R-squared		0.957	0.958		0.939	0.930
Demographics		No	No		No	No
Nationality F.E.		Yes	Yes		Yes	Yes
Year x Quarter F.E.		Yes	Yes		Yes	Yes
Pop. weights		Yes	Yes		Yes	Yes
Mean dep. var. (pre)	2.417	2.417	2.422	5.010	5.010	4.720

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.
The outcome in cols. (1-3) is the share of abortions per thousand inhabitants, while the outcome in cols. (4-6) is the share of births per thousand inhabitants. In cols. (1) and (4), we report the ATT estimates of the synthetic Difference-in-Differences, with standard errors obtained with a clustered jackknife technique; in the other columns, we report the results of a standard Difference-in-Differences, with the binary treatment cols (2) and (5), and the continuous one over the full sample (3) and (6). Standard errors are clustered at the observation level.

We next examine effects on births. Here too the direction of the effect is ambiguous: births could decline if women choose to delay childbearing, or increase if access to abortion and family planning services became more difficult. Our results support the latter. Women in the top quartile of the unauthorized share were about 9% more likely to give birth after the reform (0.46 additional deliveries per 1,000 inhabitants). The TWFE estimates are consistent: women from nationalities in the top quartile were about 12% more likely to give birth, and each additional 10 percentage points in the unauthorized share is associated with approximately 0.21 more births per 1,000 inhabitants, a 4.1% increase relative to the pre-reform mean. The Synthetic DiD results are shown in Figure 3, and dynamic estimates with jackknife standard errors in Figure A11 in the Appendix. Pre-treatment periods are

Synthetic DID estimation of abortions using the binary treatment (Top Vs Bottom quartiles in terms of estimated share of unauthorized immigrants).

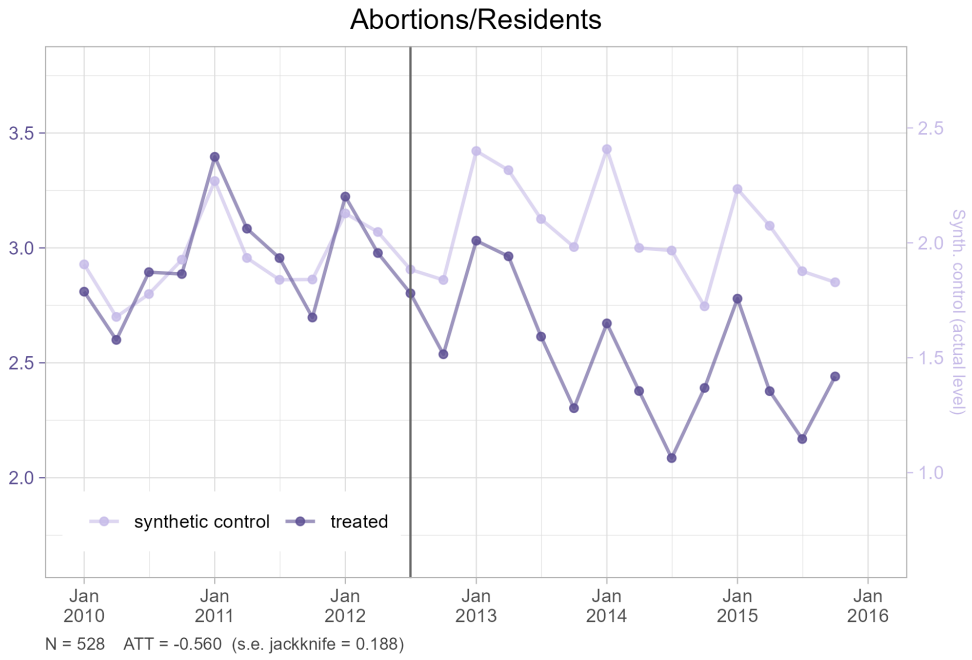


Figure 2: The plot reports the share of abortions per thousand inhabitants among women from a given nationality, in a Synthetic Difference-in-Differences design. The treated units are the countries with a share of unauthorized immigrants in the top quartile, while the control ones are the countries with a share of unauthorized immigrants in the bottom quartile. The sample includes only countries whose in the top and bottom quartiles in terms of the share of unauthorized immigrants.

well matched.

Applying the synthetic DiD to the full sample using the median as the treatment threshold yields an imprecise and small estimate (ATT =0.12, s.e. =0.196; Figure A12 in the Appendix), suggesting that only the nationalities that were most affected by the reform saw an increase in births. The standard DiD on the same binary sample (Figure A13) and the continuous treatment specification on the full sample (Figure A14) confirm the positive direction of the effect, though with less precision.

5.2 Birth-related and infant-health outcomes

We now evaluate the effect of the reform on birth-related outcomes. Beyond discouraging prenatal care utilization, the reform may have altered the composition of women who gave birth, independently affecting delivery outcomes. For these analyses, the unit of observation

Synthetic DID estimation of births using the binary treatment (Top Vs Bottom quartiles in terms of estimated share of unauthorized immigrants).

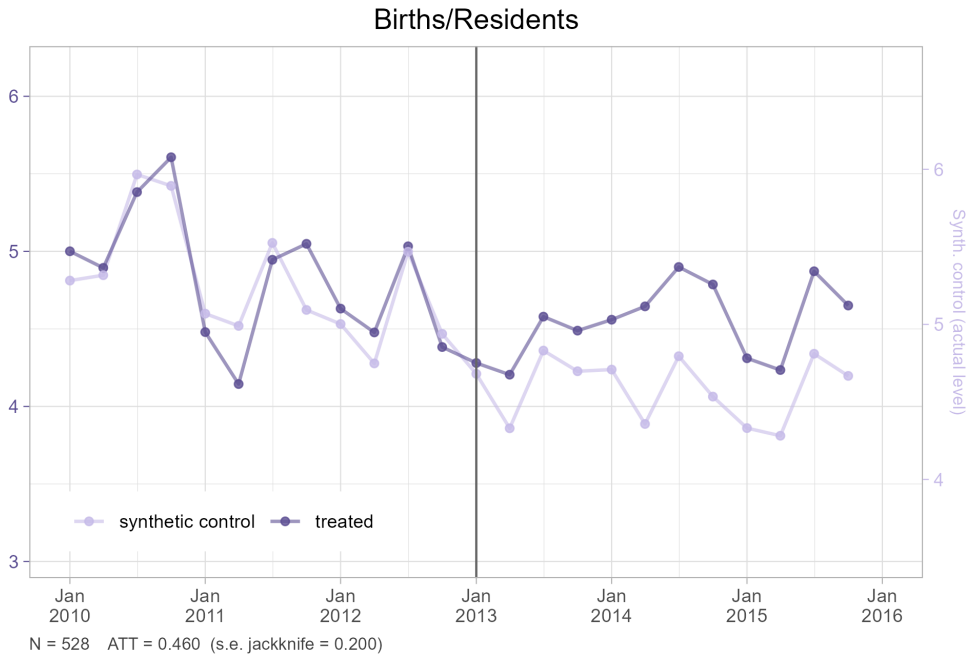


Figure 3: The plot reports the share of births per thousand inhabitants among women from a given nationality, in a Synthetic Difference-in-Differences design. The treated units are the countries with a share of unauthorized immigrants in the top quartile, while the control ones are the countries with a share of unauthorized immigrants in the bottom quartile. The sample includes only countries whose in the top and bottom quartiles in terms of the share of unauthorized immigrants.

is the individual birth, with Province x Country-of-Origin fixed effects and standard errors clustered at the same level. Outcomes include premature birth, birth complications, caesarean delivery, and low birth weight (below 2.5 kg). For comparability with the fertility results, we also aggregate outcomes at the nationality level and estimate a Synthetic DiD on shares. The static results are reported in Table 2. We prefer the individual-level TWFE specification for two reasons: it retains substantially more variation, and the pre-trends do not indicate violations of the parallel trends assumption (Figure 4). The Synthetic DiD, estimated on aggregated data, yields imprecise estimates and for some outcomes the pre-treatment periods are not well matched, making causal interpretation more challenging. We therefore treat the Synthetic DiD results as supplementary (Figure A16 in the Appendix). The TWFE estimates indicate that women in the most affected nationalities were 7% more likely to experience birth complications and 6.7% more likely to undergo a caesarean section. Effects on premature

births are negative in the TWFE but small and sensitive to specification, while effects on low birth weight are close to zero. The dynamic estimates in Figure 4 confirm these patterns and do not show pre-reform trends for complications or caesarean deliveries.

Table 2: Static Synthetic (aggregated data) and Standard Difference-in-Differences (individual data) using the binary treatment variable (estimated share of unauthorized immigrants in the top versus bottom quartile): Birth-related outcomes.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	TWFE	Synth DID	TWFE	Synth DID	TWFE	Synth DID	TWFE	Synth DID
VARIABLES	Birth prem.	Birth prem.	Weight <2.5kg	Weight < 2.5kg	Birth compl.	Birth compl.	C-section	C-section
Sh. Un. Imm.> 3Q*After	-0.008*		0.000		0.009*		0.017**	
	(0.004)		(0.005)		(0.005)		(0.007)	
After	0.003		0.000		-0.003		-0.002	
	(0.004)		(0.004)		(0.006)		(0.007)	
ATT		0.006		-0,002		0.019		0.008
		(0.007)		(0.005)		(0.012)		(0.013)
Constant	0.058***		0.058***		0.122***		0.244***	
	(0.002)		(0.003)		(0.004)		(0.005)	
Observations	286,778	528	252,257	528	286,778	528	286,778	528
Observation level	Individual	Nationality	Individual	Nationality	Individual	Nationality	Individual	Nationality
R-squared	0.007		0.008		0.019		0.015	
Demographics	No		No		No		No	
Nation. x Prov. F.E.	Yes		Yes		Yes		Yes	
Year and Month F.E.	Yes		Yes		Yes		Yes	
Mean dep. var. (pre)	0.063	0.061	0.060	0.068	0.128	0.132	0.248	0.261

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.
In all regressions, the sample includes only women whose nationality is in the top or the bottom quartile in terms of the share of unauthorized immigrants. The treatment is a binary variable flagging to which groups respondents belong to. In the odd columns, the dataset is at the country of nationality-level, so all the outcomes are in terms of shares: in col. (1), the outcome is the share of pre-term births out of the total number of births; in col. (2) it's the share of underweight babies; in col. (3), it's the share of birth complications; and in col. (4), it's the share of c-sections. We report the overall ATT, along with the standard errors calculated with a jackknife procedure. In the even columns, instead, the observations are at the individual level, so each outcome is a dummy variable =1 if the event described occurs. The coefficients reported are the interaction coefficients of a standard TWFE regression. Standard errors are clustered at the country of origin × province of residence level.

Results using the continuous treatment are reported in Figure A15 in the Appendix and are broadly consistent.

5.3 Mother's composition effect

We propose two mechanisms that may explain the birth outcome results: a composition effect, whereby the reform altered the profile of women who gave birth, and a prenatal care effect, whereby unauthorized women may have avoided seeking prenatal care due to uncertainty about their entitlements or fear of deportation. To investigate the composition effect, we examine whether the reform changed the demographic characteristics of mothers. The static results in Table 3 show that mothers from nationalities in the top quartile were 1.9 percentage points more likely to be over 35 and 3.1 percentage points less likely to have a partner, both significant in the TWFE. These effects are not robust to the Synthetic DiD, likely reflecting

Standard DID estimation of birth-related outcomes with individual-level data using the binary treatment (Top Vs Bottom quartiles in terms of estimated share of unauthorized immigrants).

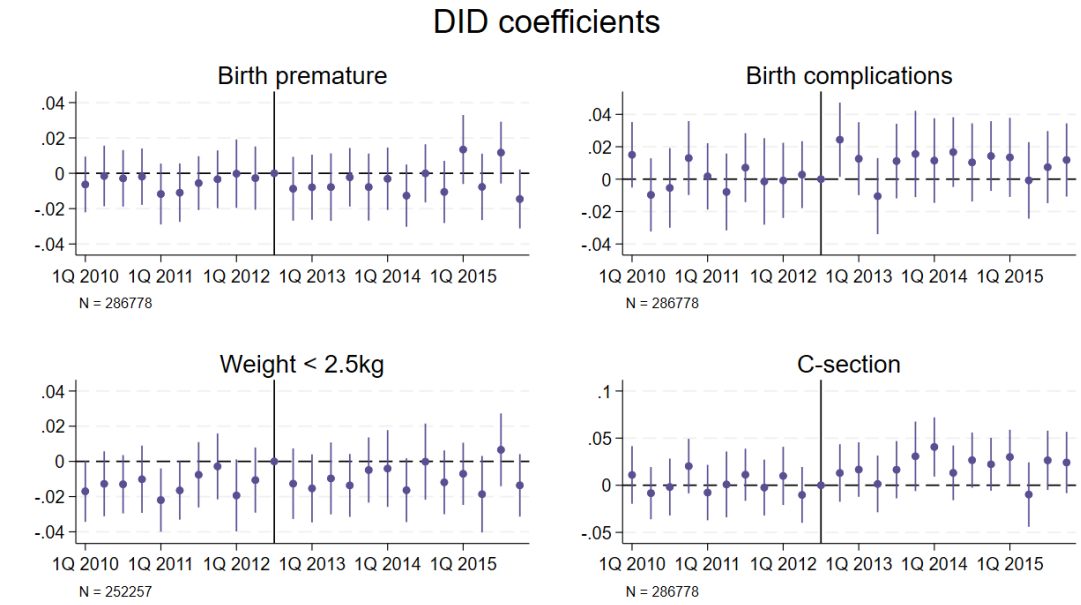


Figure 4: Interaction coefficients β_t between year quarter and a dummy = 1 if the share of unauthorized immigrants from the woman's nationality is in the bottom quartile. The sample includes only nationalities in the top and bottom quartiles. The outcome in the top left figure is a dummy variable equal to 1 if the birth was premature; the one in the top right figure flags whether the newborn was moderately underweight (<2,500 g); the one in the bottom left figure, whether there were birth complications; and the bottom right figure whether a c-section was performed.

the noisier aggregate data. We find no significant effects on teenage motherhood or missing father information. Dynamic estimates are displayed in Figure 5.

The dynamic estimates point in the same direction - older and less partnered mothers among the most affected groups - but the effects are not sharp. Results using the continuous treatment and full sample (Figure A17 in the Appendix) are consistent with this mild pattern. We also examine compositional changes in the abortion sample (Table A4, Figures A19 and A20 in the Appendix) and find no strong evidence of compositional change among women seeking abortions.

Table 3: Static Synthetic (aggregated data) and Standard Difference-in-Differences (individual data) using the binary treatment variable (estimated share of unauthorized immigrants in the top Vs bottom quartile): Mothers' characteristics in the births sample.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	TWFE	Synth DID	TWFE	Synth DID	TWFE	Synth DID	TWFE	Synth DID
VARIABLES	Mother Teenager	Share Mother Teenager	Mother >35 y.o.	Share Mother >35 y.o.	Share Single Mother	Single Mother	Father missing	Share Father missing
Sh. Un. Imm. >3Q*After	0.002		0.019***		0.031***		0.004	
	(0.002)		(0.005)		(0.009)		(0.003)	
After	-0.006***		0.026***		0.012		0.004	
	(0.002)		(0.006)		(0.009)		(0.002)	
ATT		0.002		0.013*		0.006		-0.003
		(0.004)		(0.008)		(0.020)		(0.003)
Constant	0.032***		0.169***		0.226***		0.029***	
	(0.002)		(0.004)		(0.004)		(0.001)	
Observations	286,778		286,778		286,778		286,778	
Observation level	Individual	Nationality	Individual	Nationality	Individual	Nationality	Individual	Nationality
R-squared	0.019		0.018		0.086		0.070	
Demographics	No		No		No		No	
Nation. x Prov. F.E.	Yes		Yes		Yes		Yes	
Year and Month F.E.	Yes		Yes		Yes		Yes	
Mean dep. var. (pre)	0.038	0.031	0.160	0.179	0.217	0.256	0.029	0.049

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.
Even columns report the ATT of the synthetic DID and the outcome is always in shares (over the total number of women giving birth). Odd columns report the coefficients of interest of the standard DID, and the outcomes are dummy variables =1 if the event occurs. The outcome in cols.(1)-(2) is whether the mother is a teenager; in cols. (3)-(4) whether the mother is above 35 years old; in cols. (5)-(6) whether the mother is not married, nor in a civil union; in cols. (7)-(8) whether there is no information about the father. Standard errors are clustered at the country of origin × province of residence level.

5.4 Regional implementation and additional results

Regional implementation. We examine whether the effects of the reform differ by the degree of regional implementation, following Bellés Obrero et al. (2023). We classify regions into two groups: those that offered alternative care arrangements after the reform ("low-implementation") and those that imposed strict requirements or offered no alternative ("high-implementation")[6]. We collapse births and abortions to the nationality x region of residence level and run a TWFE model interacting the treatment variable with the implementation level.

We find no substantial differences across implementation groups (Table A5 in the Appendix). Several explanations are consistent with this null result. First, the reform may have

[6]Low-implementation: Andalusia, Aragon, Asturias, Catalonia, Extremadura, Galicia, Basque Country, Navarra; high-implementation: Canary Islands, Cantabria, Castilla-La Mancha, Castilla y León, Valencia, Madrid, Murcia, La Rioja.

Standard DID estimation of the mothers' characteristics with individual-level data using the binary treatment (Top Vs Bottom quartiles in terms of estimated share of unauthorized immigrants).

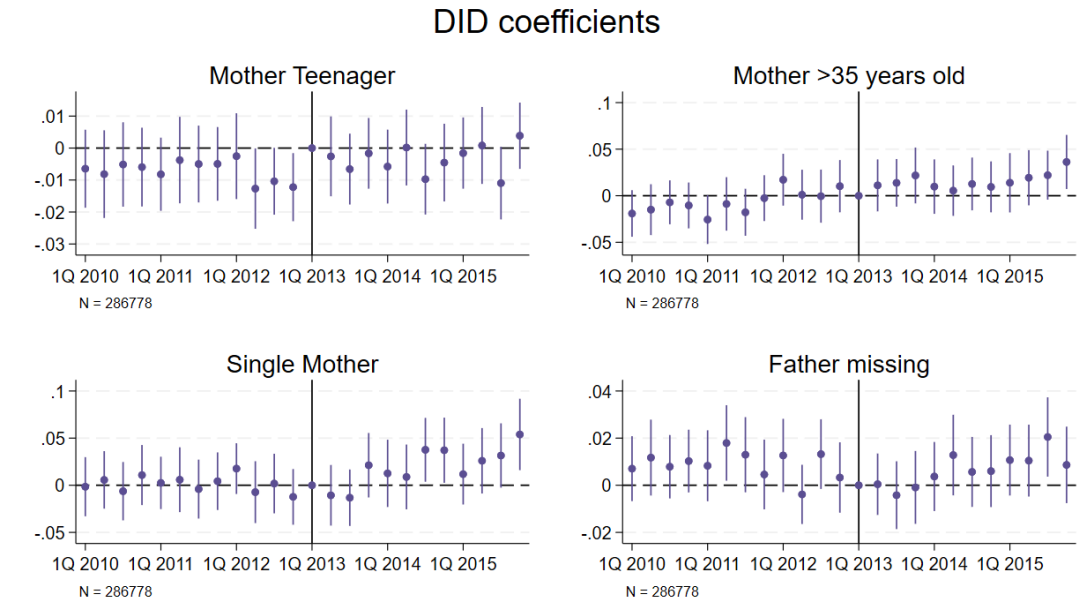


Figure 5: Interaction coefficients β_t between year quarter and the share of unauthorized immigrants from the woman's nationality from the DID regression. The outcome in the top left figure is a dummy variable equal to 1 if the mother is a teenager; the outcome in the top right figure flags whether she is older than 35; the bottom left figure whether she is not married nor in a civil union, and the bottom right figure whether the father is missing from the data. The period of reference is the first quarter of 2013, two quarters after which the reform was implemented, and after which demographic changes could be observed.

operated primarily through deterrence rather than through changes in institutional access, in which case regional variation in implementation would matter less than the mere presence of the reform itself. This interpretation is consistent with anecdotal evidence from NGOs[7]. working with immigrant communities, who reported that unauthorized immigrants became increasingly fearful of engaging with public institutions following the reform, regardless of the specific implementation status of their region. Second, data aggregation and internal migration may introduce sufficient noise to obscure genuine heterogeneity. However, we find limited support for a substantial compositional shift in this dimension: Figure A6 shows no

[7]For instance, the organizations and associations that formed REDER (Red de Denuncia y Resistencia al RDL 16/2012) reported that administrations engaged in deterrent and fear-inducing actions, and that several pregnant women were not granted access to health care. See their report here: `https://www.reder162012.org/wp-content/uploads/2022/02/REDERresumen15-ENG.pdf` (Access on July, 9^{th}2026).

clear evidence of internal migration from high- towards low-implementation regions, except possibly in the third year after the introduction of the reform, a horizon that falls after the window in which we detect effects on reproductive health outcomes. This timing further supports a deterrence-based interpretation over one driven by relocation toward areas with weaker enforcement.

Contraceptive pill usage. We investigate whether the increase in births is partly explained by reduced contraceptive use. We exploit the 2012 Spanish Health Survey (ENSE), conducted entirely before the reform, and the 2014 European Health Survey (EESE), conducted two years after, to compare self-reported pill usage among non-Spanish women aged 15-60. Given the small sample of foreign-born women (approximately 750 out of 13,500 women in the relevant age range), we compare foreign residents to native women. Overall, we find no significant decline in pill usage among foreign women. However, in the heterogeneous analysis by regional implementation intensity, non-Spanish women in high-implementation regions were about 8 percentage points less likely to report using the pill after the reform (Table A6 in the Appendix).

6 Discussion and conclusions

We provide causal evidence on the consequences of restricting healthcare access (for unauthorized immigrant women). Exploiting the 2012 Spanish reform that removed public healthcare entitlements for unauthorized immigrants, we document substantial effects on fertility behaviors and delivery outcomes. Using administrative data covering the universe of births and abortions in Spain and a synthetic difference-in-differences framework, we find that women from nationalities with a high share of unauthorized immigrants (more affected by the reform) were about 23% less likely to terminate a pregnancy and roughly 10% more likely to give birth after the reform. These magnitudes suggest that the policy significantly curtailed women's ability to control fertility, primarily by reducing access to abortion services rather than through changes in contraceptive behavior.

We also find evidence of deteriorating delivery conditions among the most affected groups, with higher rates of birth complications and caesarean deliveries. The evidence on neonatal health indicators is more mixed. In part, these effects likely reflect composition changes—after the reform, mothers from the most affected nationalities were more likely to be over 35 and less likely to have a partner, both characteristics associated with higher delivery risk. Reduced access to prenatal care may have compounded these effects, though we cannot fully disentangle the two mechanisms.

Our findings contribute to a growing literature on the unintended consequences of restrictive healthcare policies. We show that even in a high-income country with a universal healthcare system and formal exemptions for pregnancy-related care, exclusionary policies can substantially disrupt fertility control and worsen delivery outcomes among vulnerable populations. This distinguishes our setting from most existing evidence, which comes from the United States or developing countries, and suggests that the consequences of healthcare exclusion are not confined to fragmented or resource-constrained systems.

Several questions remain for future research. A key open issue is the extent to which prenatal care utilization declined among affected women and how this contributed to the observed deterioration in delivery outcomes. The longer-term consequences for children born during the period of restricted access also remain to be explored.

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7 Appendix

A1.1 Descriptive evidence

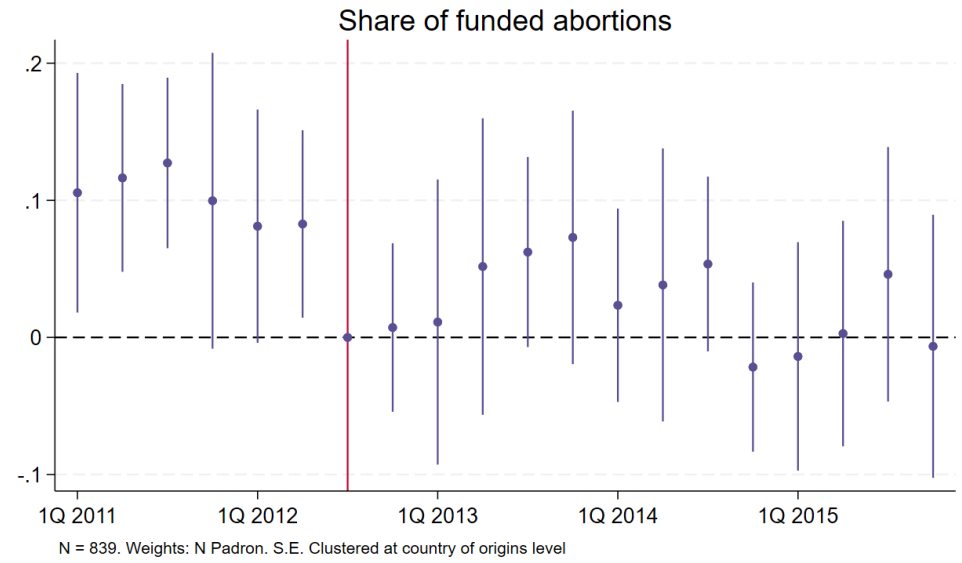


Figure A1: Interaction coefficients β_t between year quarter and the share of unauthorized immigrants from the woman's nationality. The outcome is the share of abortions whose cost was covered by the state. This information was available only starting from 2011.

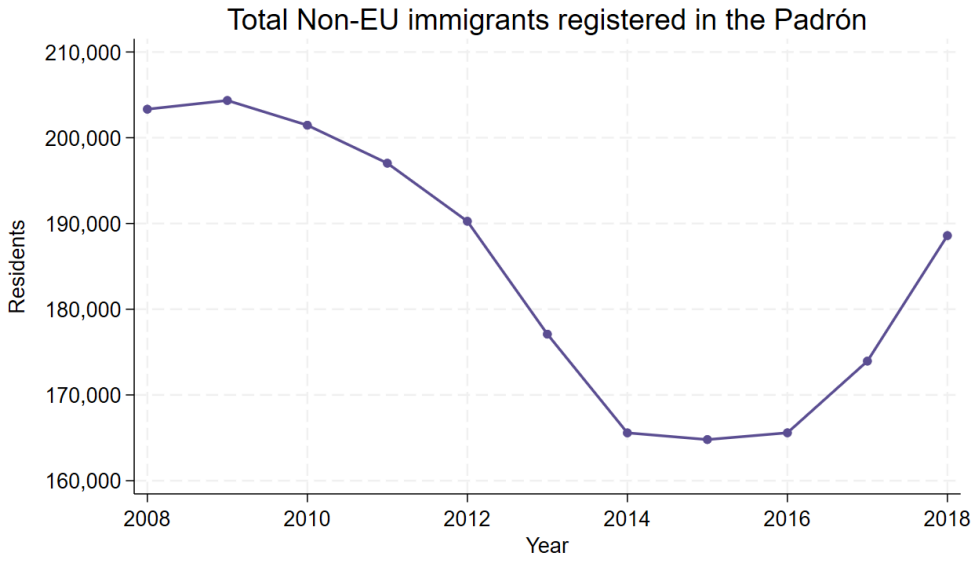


Figure A2: Total number of Padron registered by immigrants from countries not covered by the Schengen agreements in time

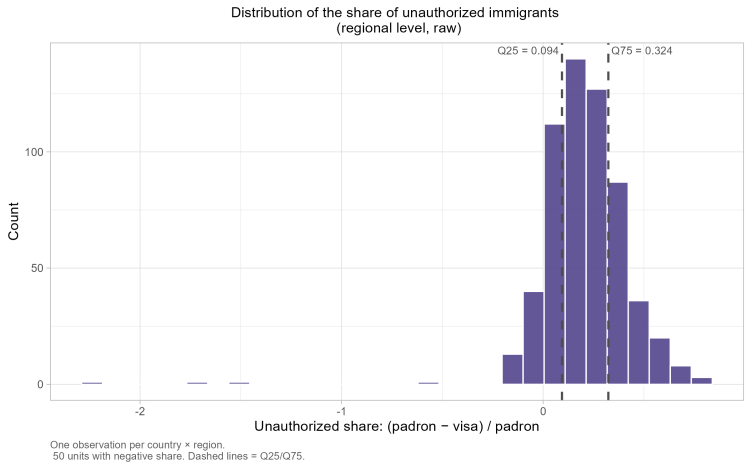


Figure A4: Histogram reporting the fraction of unauthorized immigrants from the women's country of origin, at the regional level. The figure reports the percentage of countries with the share of unauthorized immigrants in 2011 in each bin.

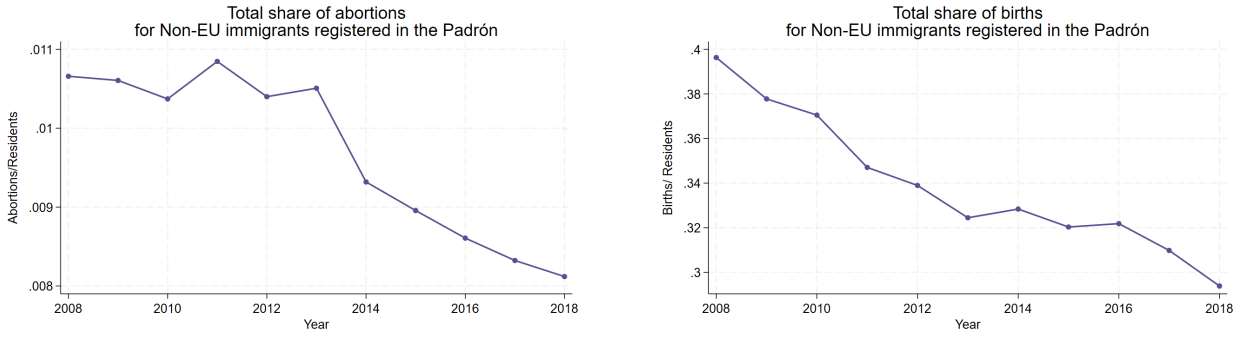


Figure A3: Share of Non-Schengen immigrant women undergoing an abortion (left) and giving birth (right)

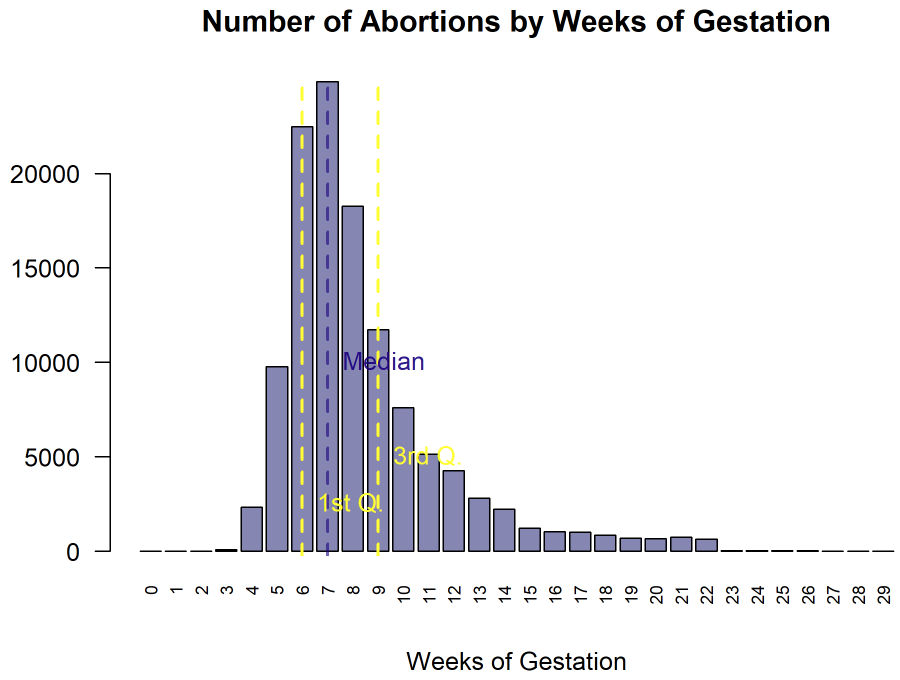


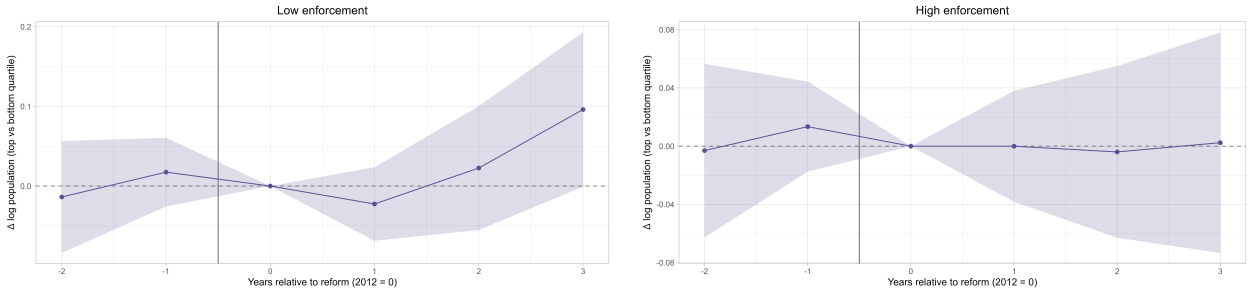
Figure A5: Number of abortions by week of gestation in 2011

A1.2 Methodology

A1.2.1 Share of unauthorized immigrants

Table A1: Share of unauthorized immigrants by country of origin, in 2011

Country	N Births	Padron	Share unauthorized Immigrants	Share <1st Quartile	Country	N Births	Padron	Share unauthorized Immigrants	Share >4th Quartile
Morocco	24722	788.563	0,023	1	Bolivia	3799	186.018	0,233	0
Ecuador	5339	308.174	0,055	1	Cape Verde	50	3.249	0,239	0
China	4584	177.001	0,055	1	Senegal	899	63.963	0,249	0
Philippines	534	30.641	0,085	1	Uruguay	507	38.980	0,249	0
Peru	2130	122.643	0,091	1	Bangladesh	232	13.075	0,264	0
Algeria	1437	62.912	0,107	1	Argentina	1658	109.258	0,268	0
Gambia	448	21.984	0,109	1	Guinea	237	12.417	0,291	0
Moldova	335	17.853	0,112	1	Nigeria	1568	46.296	0,299	0
Mauritania	187	11.510	0,123	1	Armenia	171	11.852	0,303	0
Japan	130	5.465	0,133	1	Angola	39	1.989	0,313	0
Colombia	3968	246.345	0,143	1	Cameroon	113	6.360	0,314	1
Ukraine	1077	88.966	0,148	0	Chile	541	39.204	0,319	1
India	591	35.888	0,172	0	United States	358	27.856	0,341	1
Belarus	86	3.906	0,177	0	Russia	1067	57.644	0,343	1
Dominican Republic	1783	92.846	0,177	0	Georgia	156	11.629	0,361	1
Ghana	245	16.971	0,178	0	Venezuela	982	59.322	0,373	1
Syria	70	2.575	0,194	0	Brazil	2069	99.870	0,400	1
Mali	315	24.634	0,196	0	Equatorial Guinea	410	14.374	0,408	1
Cuba	741	54.958	0,198	0	Mexico	535	25.567	0,427	1
Guinea-Bissau	73	5.398	0,206	0	Paraguay	1923	88.327	0,536	1
Pakistan	1550	79.984	0,224	0	Honduras	719	36.016	0,579	1



(a) Low Implementation (b) High Implementation

Figure A6: Event-study estimates of the log (number of residents) for the top (vs bottom) quartiles of the probability of being unauthorized, by regional implementation of the law.

Table A2: Demographic characteristics of the women of a nationality not part of the Schengen agreements who had a voluntary interruption of pregnancy in Spain in 2011

Variable	**Mean**	**SD**	**N**
Woman's age	28,744	6,559	34194
Teen woman	0,089	0,285	34194
Woman> 35y.o.	0,166	0,372	34194
Woman illeterate	0,032	0,177	33105
Woman less HS	0,641	0,480	33105
Woman HS	0,255	0,436	33105
Woman Bachelor	0,072	0,259	33105
City< 10k	0,072	0,258	34051
City 10-50k	0,528	0,499	34051
City 50-500k	0,401	0,490	34051
City> 500k	0,345	0,475	34051
Woman has children	0,672	0,470	34195
Tot children	1,243	1,197	34195
Tot abortions	0,647	0,940	34195
Weeks gestation	8,351	3,051	34195
Lives with partner	0,566	0,496	32868
Woman single	0,434	0,496	32868
Woman has income	0,604	0,489	33466
Woman works	0,567	0,496	33742
First abortion	0,562	0,496	34195
Uses contraceptive	0,498	0,500	24553

Table A3: Demographic characteristics of the women of a nationality not part of the Schengen agreements who gave birth in Spain in 2011

Variable	Mean	SD	N
Mother's age	29,35	5,75	68378
Mother Teenager	0,04	0,19	68378
Mother >35 years old	0,15	0,36	68378
Mother is illiterate	0,19	0,39	62885
Mother has no High School	0,46	0,50	62885
Mother has High School	0,17	0,38	62885
Mother has bachelor	0,17	0,38	62885
City of delivery: Capital	0,48	0,50	68373
City of delivery: >100k people	0,18	0,38	68373
City of delivery: 20-100k people	0,29	0,45	68373
City of delivery: <20k people	0,05	0,22	68378
First delivery	0,44	0,50	68378
Twins delivery	0,03	0,16	68378
Delivery at home	0,00	0,05	68378
Delivery assisted	1,00	0,03	68378
Father foreigner	0,77	0,42	66117

A1.3 Results

A1.3.1 Main

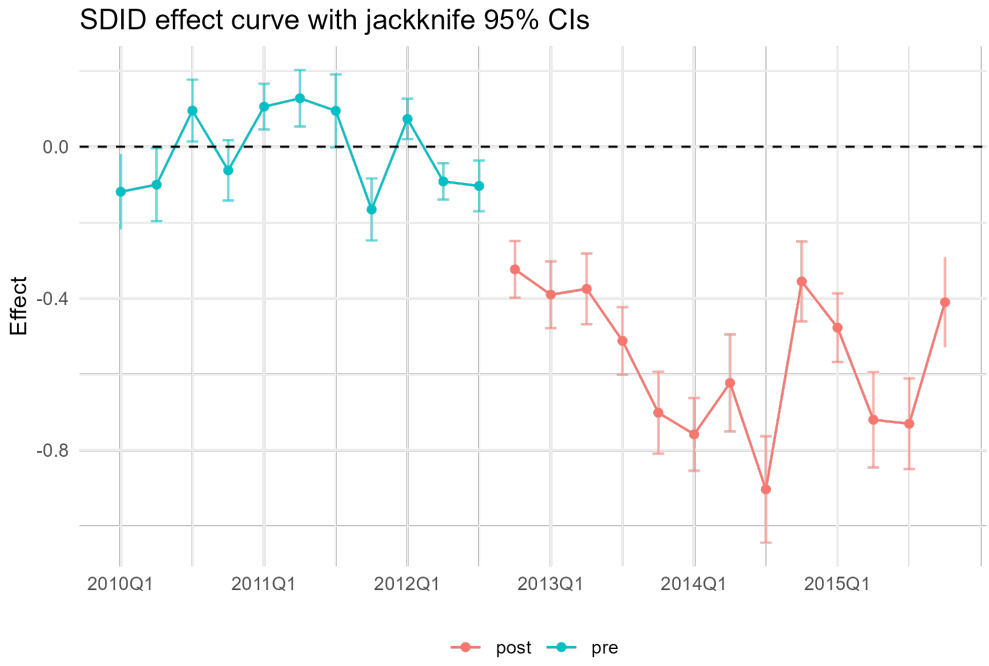


Figure A7: The plot reports the ATT in terms of share of abortions per thousand inhabitants among women from a given nationality, of the Synthetic Difference-in-Differences regression reported in 2. The treated units are the countries with a share of unauthorized immigrants $> 3^{rd}$ quartile, while the control ones are the countries with a share of unauthorized immigrants $< 1^{st}$ quartile. Standard errors are estimated through a jackknife procedure.

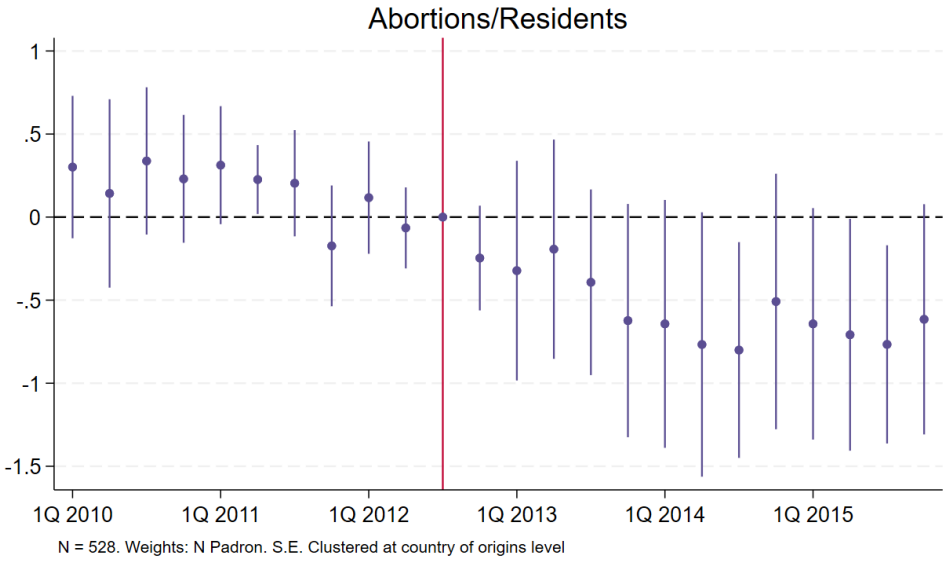


Figure A9: Interaction coefficients β_t between year quarter and a dummy = 1 if the share of unauthorized immigrants from the woman's nationality is in the bottom quartile. The sample includes only nationalities in the top and bottom quartiles. The outcome is the share of abortions per thousand inhabitants among women from a given nationality.

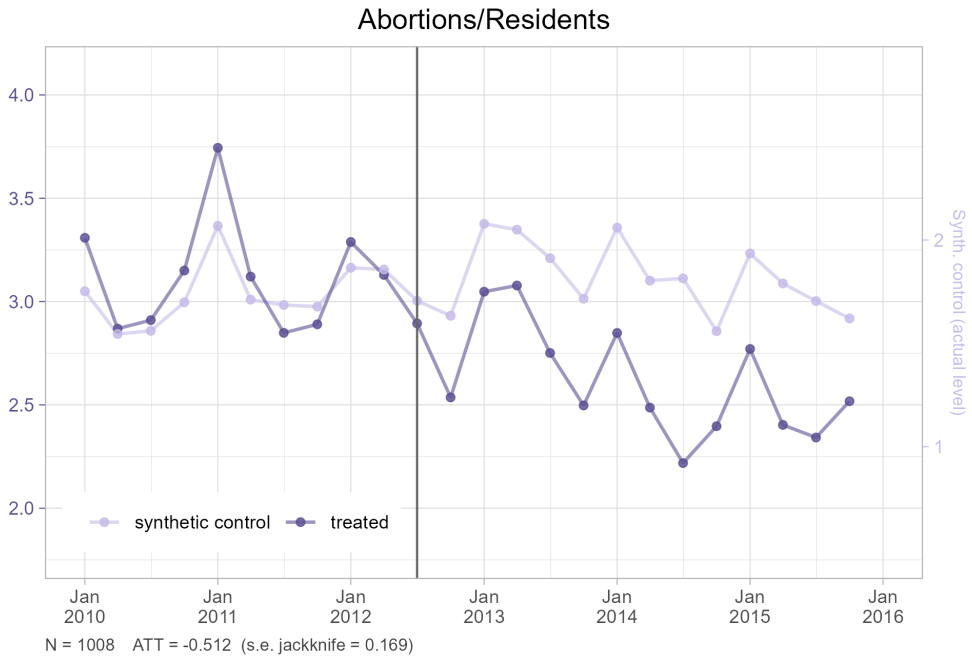


Figure A8: The plot reports the share of abortions per thousand inhabitants among women from a given nationality, in a Synthetic Difference-in-Differences design. The treated units are the countries with a share of unauthorized immigrants above the median, while the control ones are the countries with a share of unauthorized immigrants below.

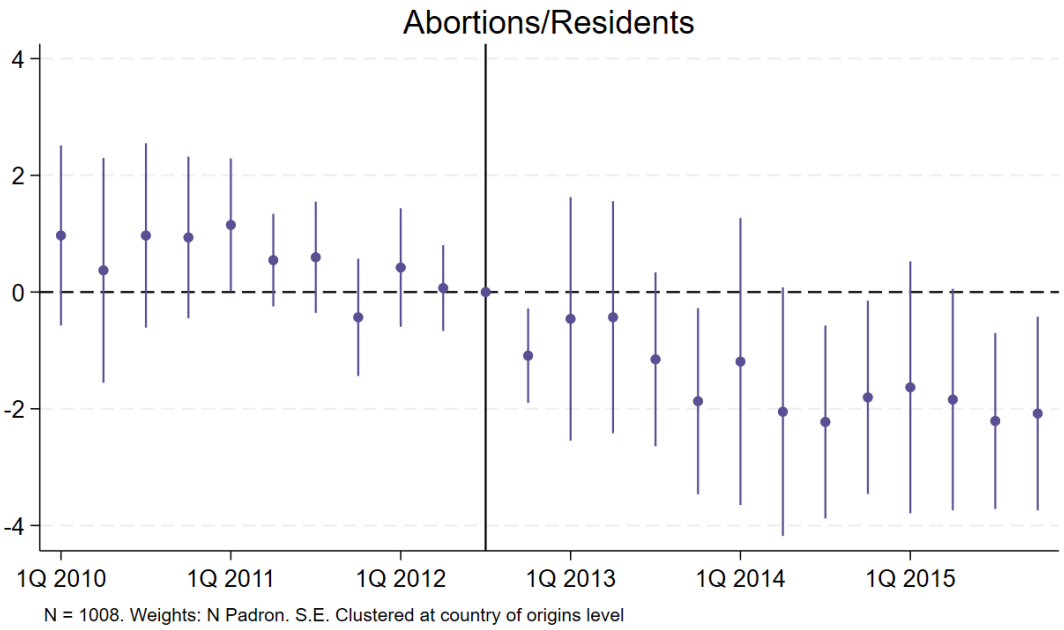


Figure A10: Interaction coefficients β_t between year quarter and the share of unauthorized immigrants from the woman's nationality from the DID regression. The outcome is the share of abortions per thousand inhabitants among women from a given nationality. The period of reference is the third quarter of 2012, in which the reform was approved.

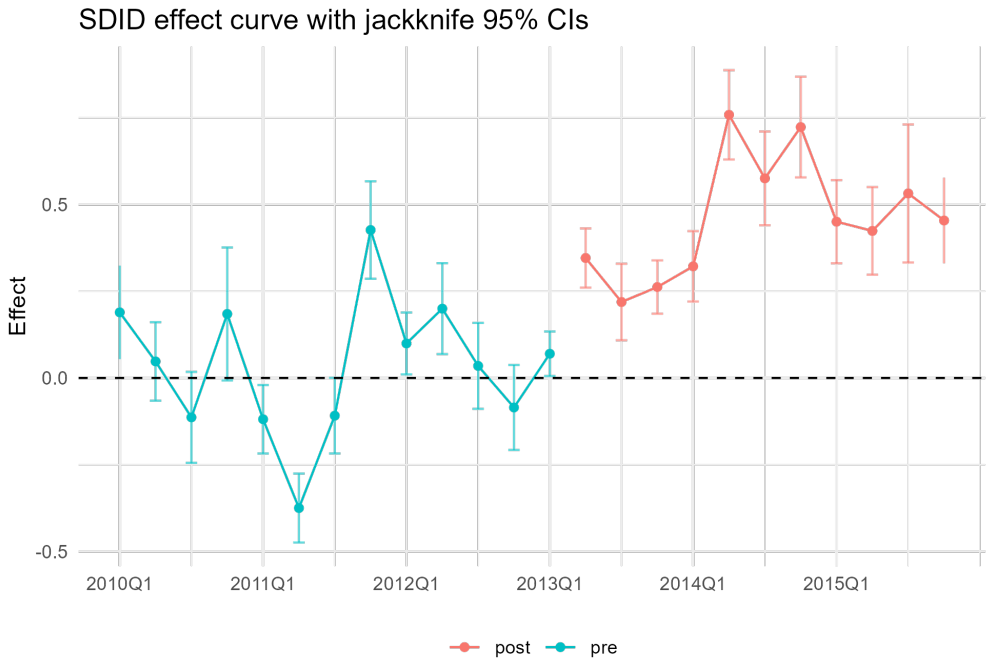


Figure A11: The plot reports the ATT in terms of share of births per thousand inhabitants among women from a given nationality, of the Synthetic Difference-in-Differences regression reported in 3. The treated units are the countries with a share of unauthorized immigrants $> 3^{rd}$ quartile, while the control ones are the countries with a share of unauthorized immigrants $< 1^{st}$ quartile. Standard errors are estimated through a jackknife procedure.

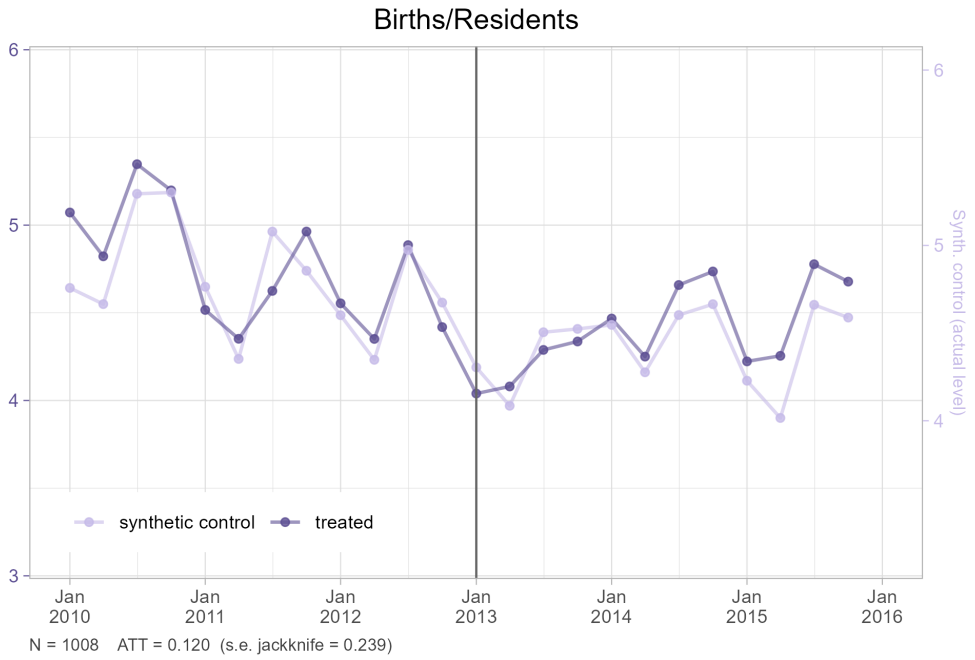


Figure A12: The plot reports the share of abortions per thousand inhabitants among women from a given nationality, in a Synthetic Difference-in-Differences design. The treated units are the countries with a share of unauthorized immigrants above the median, while the control ones are the countries with a share of unauthorized immigrants below.

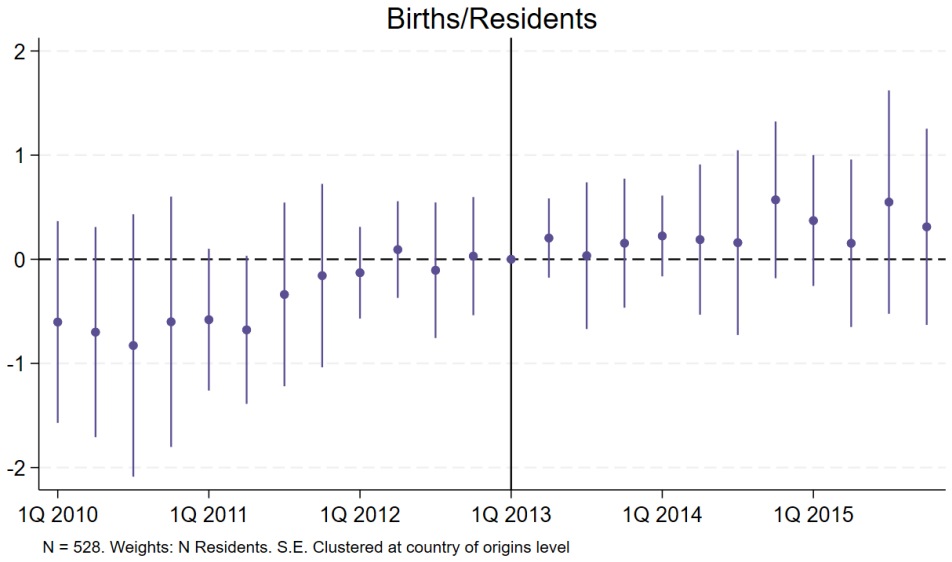


Figure A13: Interaction coefficients β_t between year quarter and a dummy = 1 if the share of unauthorized immigrants from the woman's nationality is in the bottom quartile. The sample includes only nationalities in the top and bottom quartiles. The outcome is the share of births per thousand inhabitants among women from a given nationality.

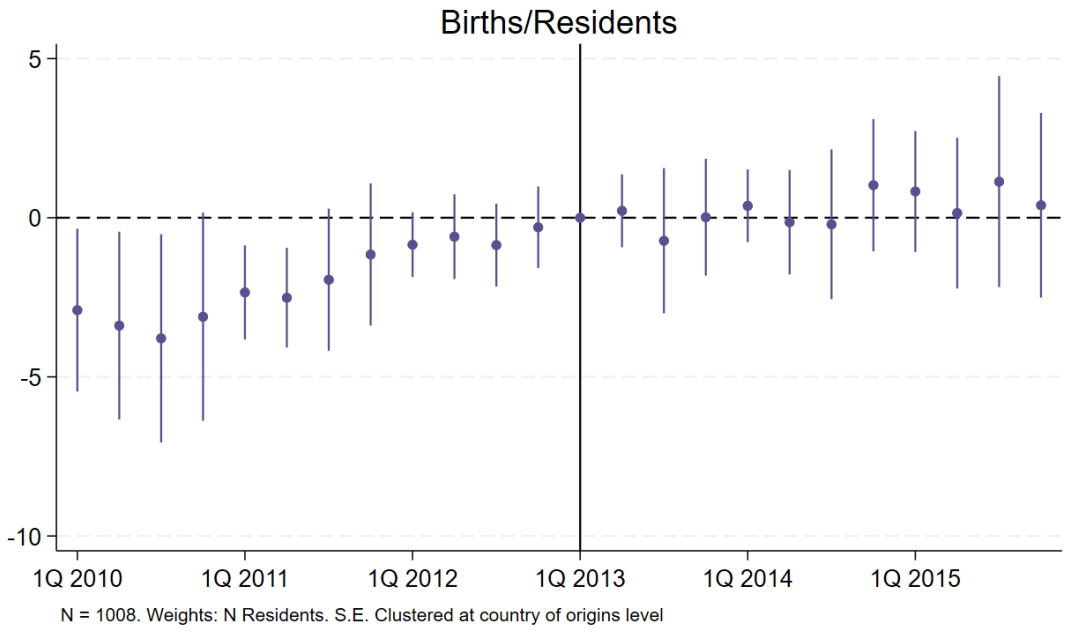


Figure A14: Interaction coefficients β_t between year quarter and the share of unauthorized immigrants from the woman's nationality from the DID regression. The outcome is the share of births per thousand inhabitants among women from a given nationality. The period of reference is the third quarter of 2012, in which the reform was approved.

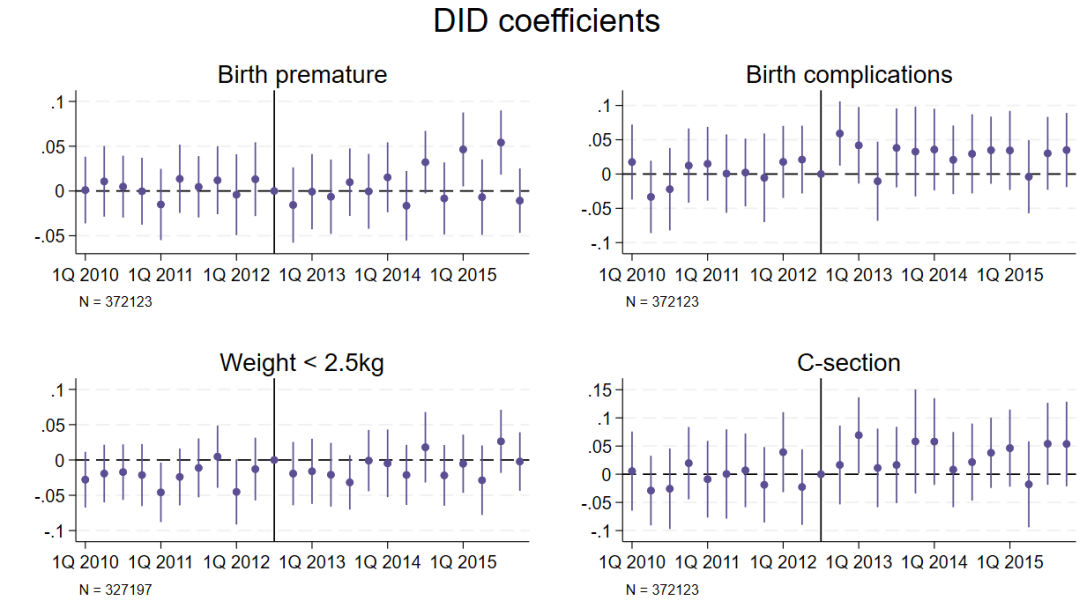


Figure A15: Interaction coefficients β_t between year quarter and the share of unauthorized immigrants from the woman's nationality from the DID regression. The outcome in the top left figure is a dummy variable equal to 1 if the birth was premature; the one in the top right figure flags whether the newborn was moderately underweight (<2,500 g); the one in the bottom left figure, whether there were birth complications; and the bottom right figure, whether a c-section was performed. The period of reference is the third quarter of 2012, in which the reform was approved.



Figure A16: The plot reports the share of women displaying some characteristics out of all the women giving birth by nationality, in a Synthetic Difference-in-Differences design. The treated units are the countries with a share of unauthorized immigrants $> 3^{rd}$ quartile, while the control ones are the countries with a share of unauthorized immigrants $< 1^{st}$ quartile. The outcome in the top left figure is a dummy variable equal to 1 if the birth was premature; the one in the top right figure flags whether the newborn was moderately underweight (<2,500 g); the one in the bottom left figure, whether there were birth complications; and the bottom right figure whether a c-section was performed. The period of reference is the third quarter of 2012, in which the reform was approved.

A1.3.2 Mother's composition effects

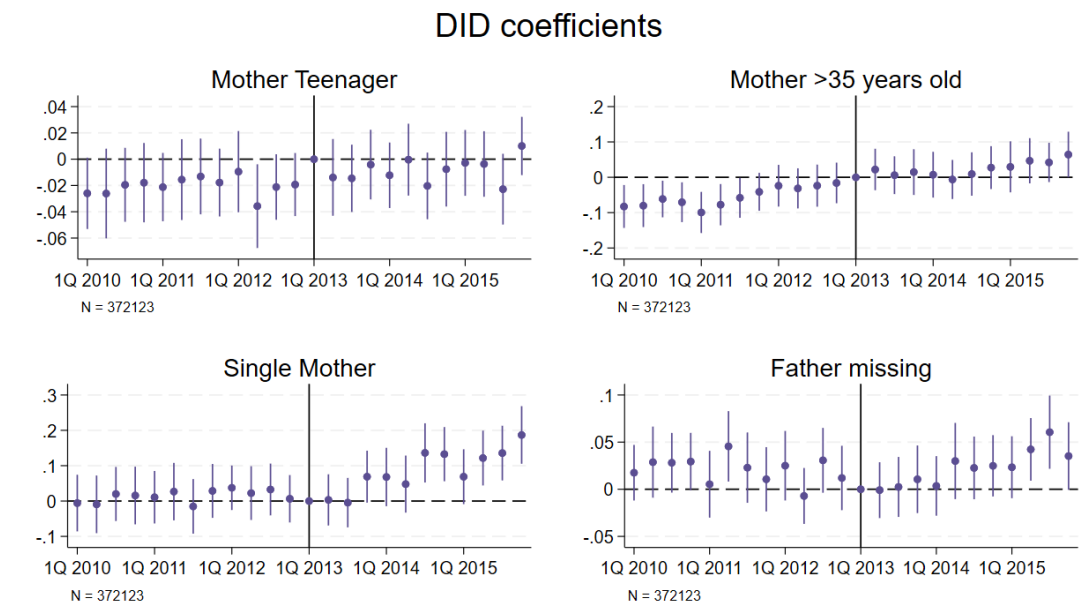


Figure A17: Interaction coefficients β_t between year quarter and the share of unauthorized immigrants from the woman's nationality from the DID regression. The outcome in the top left figure is a dummy variable equal to 1 if the mother is a teenager; the outcome in the top right figure flags whether she is older than 35; the bottom left figure whether she is not married nor in a civil union, and the bottom right figure whether the father is missing from the data. We plot the β_t coefficients of the interaction term between year quarter and a dummy = 1 if the share of unauthorized immigrants from the mother's nationality is above the fourth quartile. The sample includes only women whose nationality is in the first or fourth quartile.

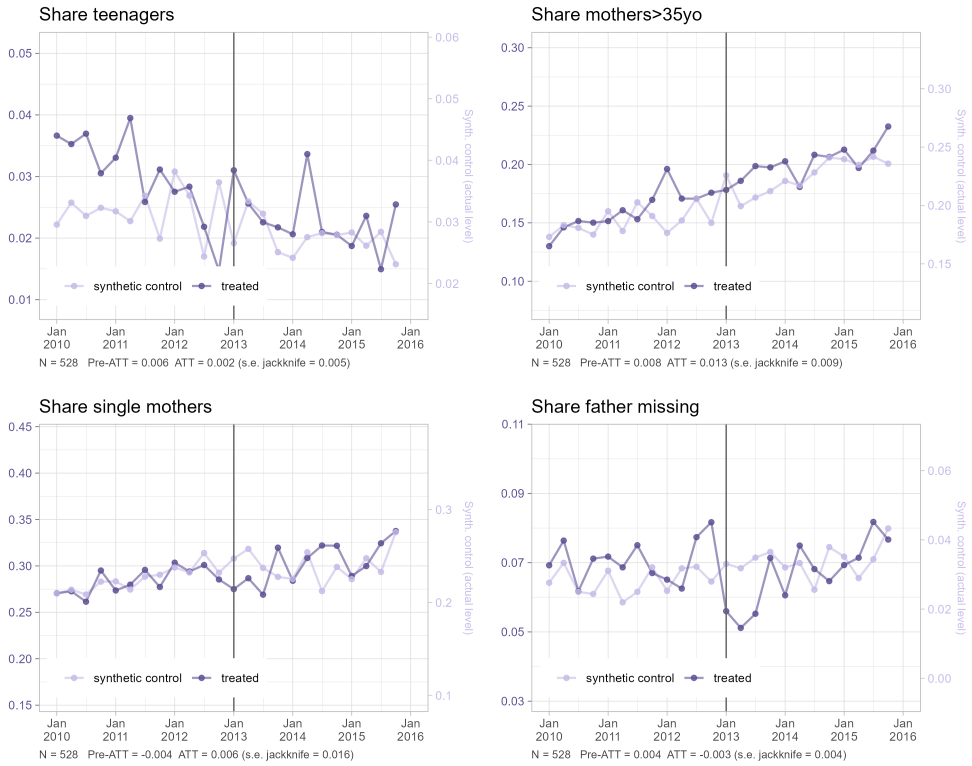


Figure A18: The plot reports the share of women displaying some characteristics out of all the women giving birth by nationality, in a Synthetic Difference-in-Differences design. The treated units are the countries with a share of unauthorized immigrants $> 3^{rd}$ quartile, while the control ones are the countries with a share of unauthorized immigrants $< 1^{st}$ quartile. The outcome in the top left figure is a dummy variable equal to 1 if the mother is a teenager; the outcome in the top right figure flags whether she is older than 35; the bottom left figure whether she is not married nor in a civil union, and the bottom right figure whether the father is missing from the data. The period of reference is the first quarter of 2013, two quarters after which the reform was implemented, and after which demographic changes could be observed.

Table A4: Static Synthetic (aggregated data) and Standard Difference-in-Differences (individual data) using the binary treatment variable (estimated share of unauthorized immigrants in the top versus bottom quartile): Women's demographic characteristics in abortion sample.

VARIABLES	(1) Share Women Teenager	(2) Woman Teenager	(3) Share Women >35 y.o.	(4) Woman >35 y.o.	(5) Share Live with partner	(6) Lives with partner
ATT	-0,006		0.01		-0.02*	
	(0.009)		(0.008)		(0.011)	
Sh. Un. Imm.>3Q*After		0.0141***		-0.0012		-0.0054
		(0.0047)		(0.008)		(0.0108)
After		-0.0208***		0.0359***		-0.0025
		(0.0081)		(0.01)		(0.0109)
Constant		0.108***		0.191***		0.519***
		(0.0054)		(0.005)		(0.0081)
Observations	528	117,365	528	117,365	528	115,365
R-squared		0.044		0.024		0.061
Demographics		No		No		No
Nationality x Province F.E.		Yes		Yes		Yes
Year and Month F.E.		Yes		Yes		Yes
Mean dep. var.		0.0993		0.188		0.532

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.
Odd columns report the ATT of the synthetic DID and the outcome is always in shares (over the total number of women giving birth). Even columns report the coefficients of interest of the standard DID, and the outcomes are dummy variables =1 if the event occurs. The outcome in cols.(1)-(2) is whether the mother is a teenager; in cols. (3)-(4) whether the mother is above 35 years old; in cols. (5)-(6) whether the mother lives with a partner or not.

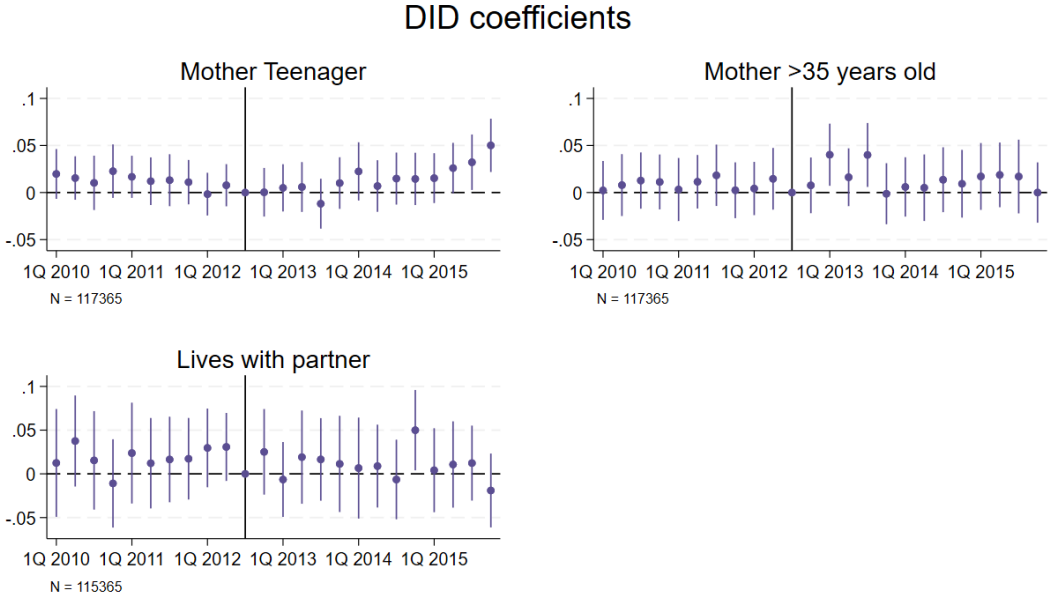


Figure A19: Interaction coefficients β_t from the DID regression. The outcome in the top left figure is a dummy variable equal to 1 if the woman having an abortion is a teenager; the outcome in the top right figure flags whether she is older than 35; and the bottom left figure whether she lives with a partner. We plot the β_t coefficients of the interaction term between year quarter and the share of unauthorized immigrants from the women's nationality.

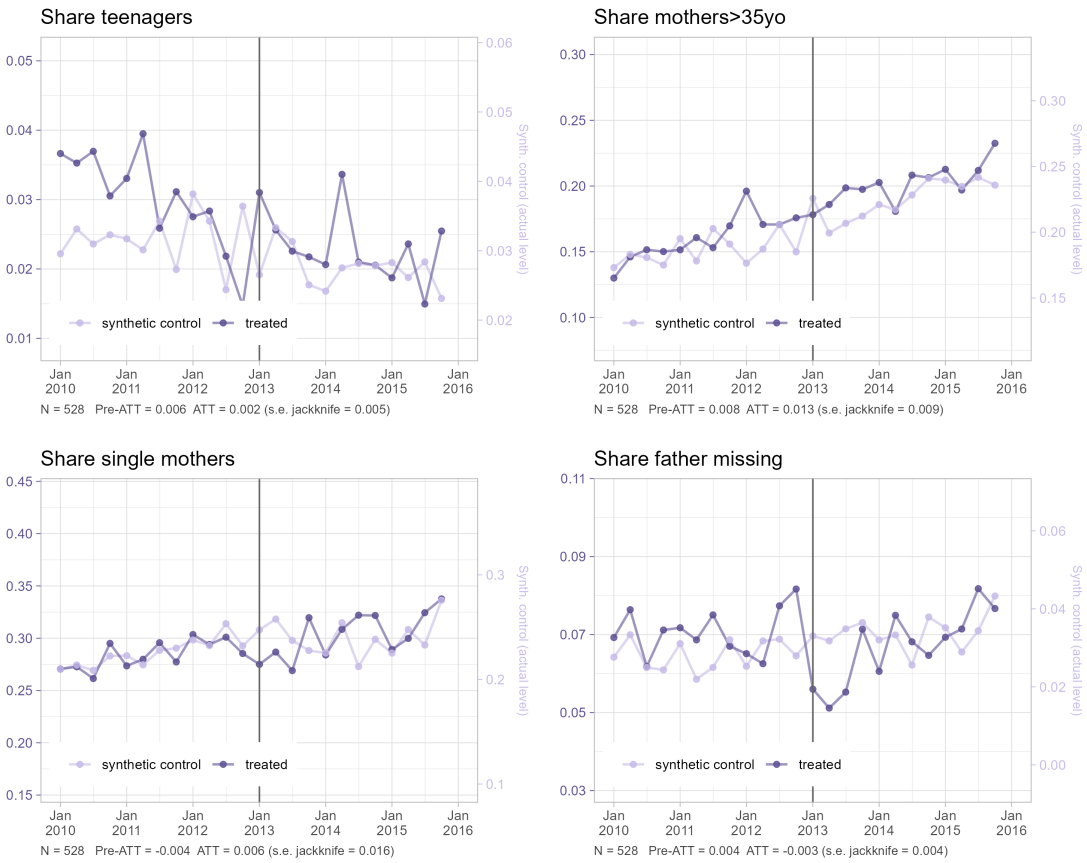


Figure A20: The plot reports the share of women displaying some characteristics out of all the women giving birth by nationality, in a Synthetic Difference-in-Differences design. The treated units are the countries with a share of unauthorized immigrants $> 3^{rd}$ quartile, while the control ones are the countries with a share of unauthorized immigrants $< 1^{st}$ quartile. The outcome in the top left figure is a dummy variable equal to 1 if the mother is a teenager; the outcome in the top right figure flags whether she is older than 35; the bottom left figure, whether she lives with a partner. We plot the β_t coefficients of the interaction term between year quarter and a dummy = 1 if the share of unauthorized immigrants from the mother's nationality is above the fourth quartile. The sample includes only women whose nationality is in the first or fourth quartile.

A1.4 Heterogeneity

A1.4.1 Regional Implementation

Table A5: Difference-in-Differences (aggregated data): Heterogeneous effects on abortions and births per capita by regional implementation of the law.

VARIABLES	(1) $\frac{N\ Abortions}{Residents}$	(2) $\frac{N\ Abortions}{Residents}$	(3) $\frac{N\ Births}{Residents}$	(4) $\frac{N\ Births}{Residents}$
Share Un. Imm.*After	-0.5618***		0.7276***	
	(0.1835)		(0.1146)	
Sh. Un. Imm.*After*High Impl.	0.1725		0.0278	
	(0.1323)		(0.0927)	
Share Un. Imm. >Q3*After		-0.2437***		0.2639***
		(0.0858)		(0.0468)
Sh. Un. Imm. >Q3*After*High Impl.		0.0649		0.0485
		(0.0551)		(0.0530)
Observations	14280	6918	16344	8232
R-squared	0.805	0.812	0.781	0.801
Nation. x Region F.E.	Yes	Yes	Yes	Yes
Year x Quarter F.E.	Yes	Yes	Yes	Yes
Pop. weights	Yes	Yes	Yes	Yes
Mean dep. var.	0.7934	0.7342	1.5907	1.7890

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.
The outcome in cols. (1-2) is the share of abortions per thousand inhabitants, while the outcome in cols. (3-4) is the share of births per thousand inhabitants. The reported coefficients are from a Difference-in-Differences specifications, with the binary treatment cols (1) and (3), and the continuous one over the full sample (2) and (4). Standard errors are clustered at the nationality x region level.

A1.4.2 Contraceptive Pill

Table A6: Contraceptive pill usage among Spanish and non-Spanish women

VARIABLES	(1) Takes contrac. pill	(2) Takes contrac. pill	(3) Takes contrac. pill	(4) Takes contrac. pill
Foreign*After	-0.0103	0.0113	0.0461*	0.0603**
	(0.0166)	(0.0159)	(0.0262)	(0.0250)
Foreign*After*High impl.			-0.0921***	-0.0788**
			(0.0336)	(0.0321)
Foreign	0.0509***	-0.00225	0.0107	-0.0450**
	(0.0115)	(0.0111)	(0.0185)	(0.0177)
After	0.0228***	-0.00726*	0.0211***	-0.00653
	(0.00393)	(0.00411)	(0.00499)	(0.00504)
High impl.*After			0.00335	-0.00151
			(0.00605)	(0.00577)
Foreign*High impl.			0.0641***	0.0682***
			(0.0232)	(0.0221)
Constant	0.0386***	0.413***	0.0386***	0.413***
	(0.00251)	(0.0161)	(0.00251)	(0.0161)
Observations	13,512	13,512	13,512	13,512
R-squared	0.005	0.096	0.006	0.097
Controls	No	Yes	No	Yes
Mean dep. var.	0.0505	0.0505	0.0505	0.0505

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.
All observations are at the individual level, so each outcome is a dummy variable equal to 1 if the respondent declared to use the contraceptive pill. The coefficients reported are the interaction coefficients of a standard two-way fixed effects regression.